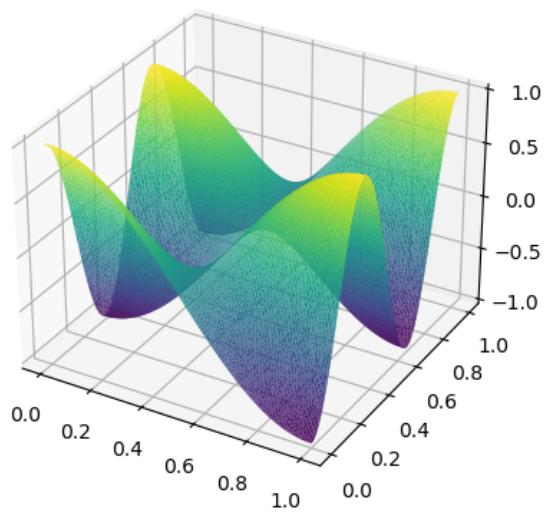
MATH 610 PA 2 Report

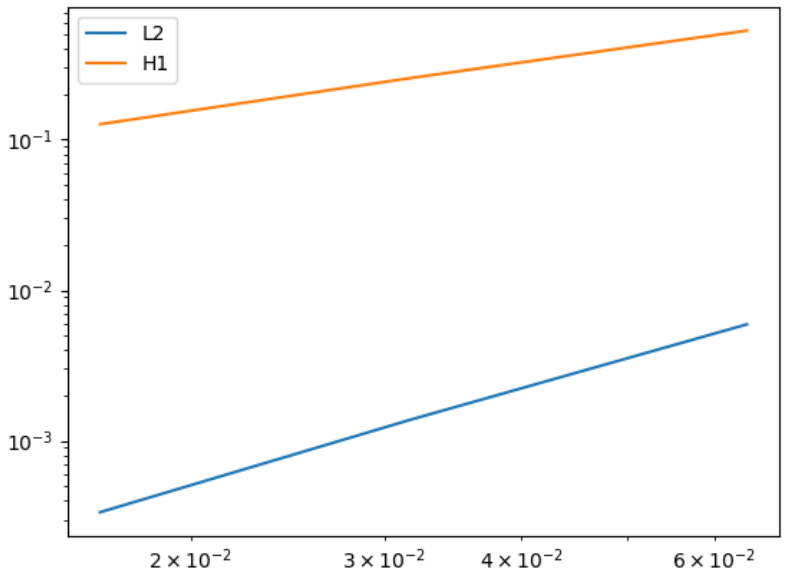
Jordan Hoffart

Problem 1

Solution



Errors



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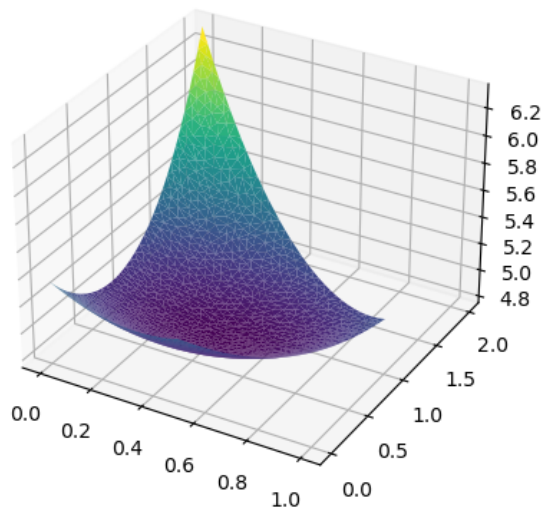
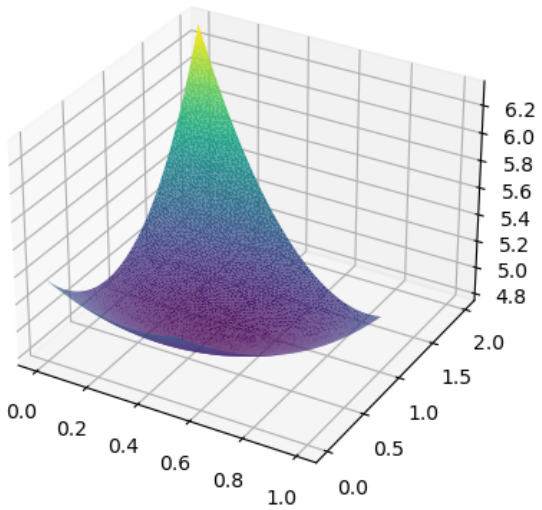
3.444E+03	3.161E-02	1.372E-03	2.064E+00	2.556E-01	1.027E+00
1.364E+04	1.652E-02	3.357E-04	2.169E+00	1.266E-01	1.082E+00

Problem 2

$q = 1$

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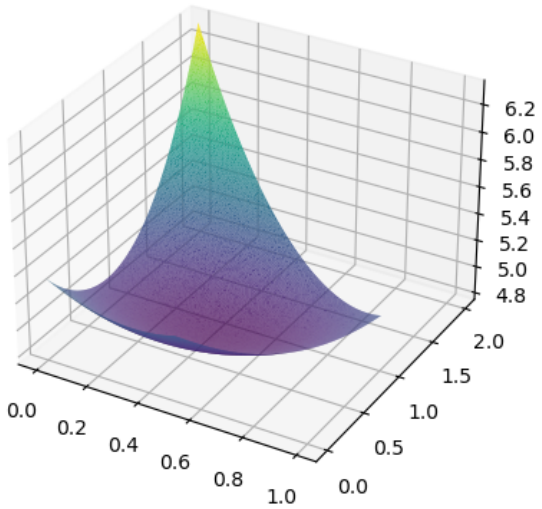
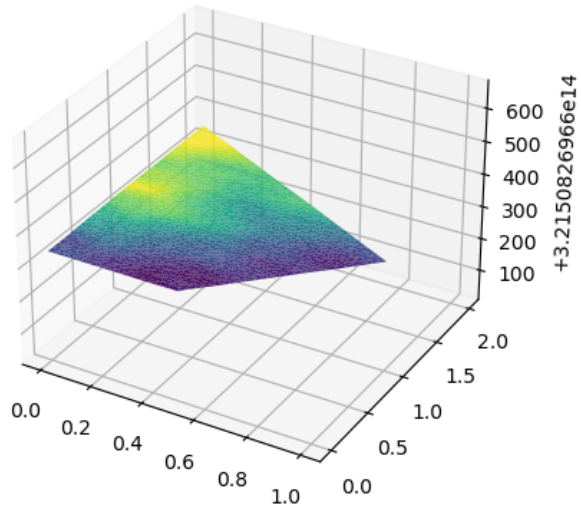
5.8
5.6
5.4
5.2
5.0
4.8
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0


6.2
6.0
5.8
5.6
5.4
5.2
5.0
4.8
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0

$q = 0$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200
100
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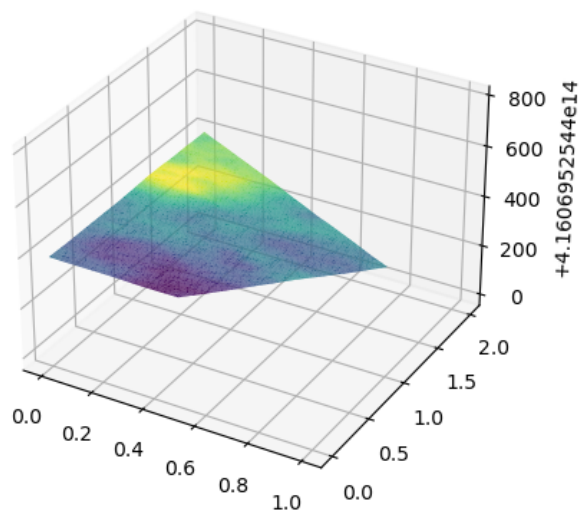
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0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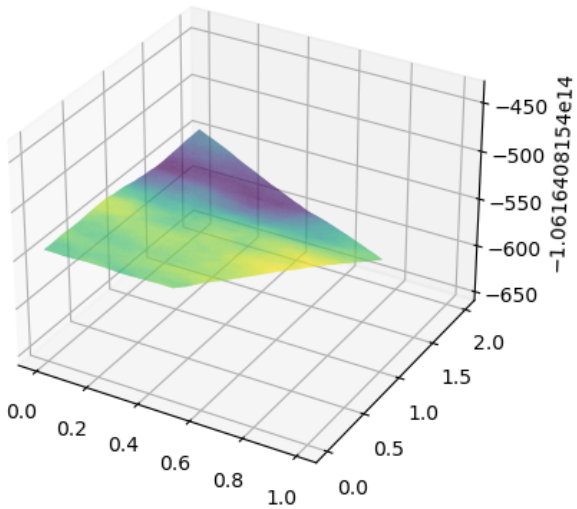
1.0
1.5
2.0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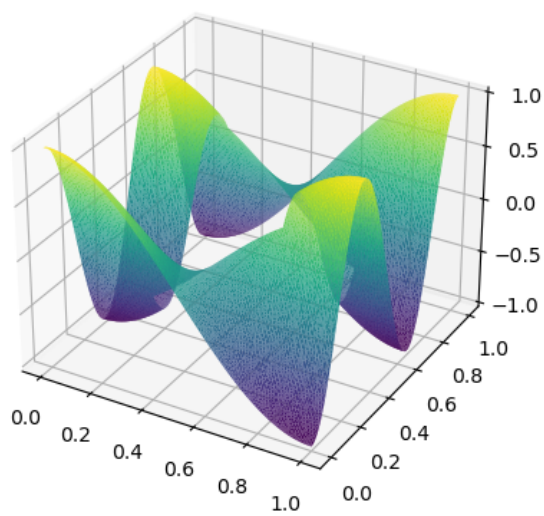
1.0
1.5
2.0

Problem 3

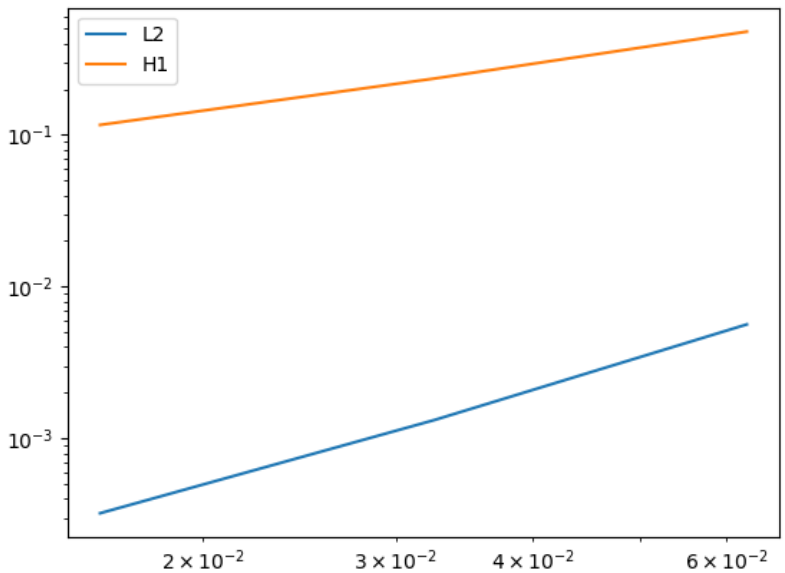
Manufactured solution

$$u(x, y) = \cos(\pi x)\cos(3\pi y)$$

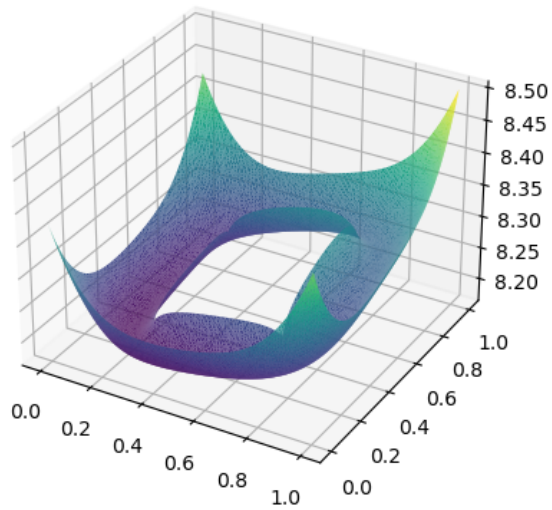
Solution



Errors



dofs	h	L2	rateL2	H1	rateH1
7.100E+02	6.265E-02	5.644E-03	0.000E+00	4.781E-01	0.000E+00
2.675E+03	3.252E-02	1.325E-03	2.211E+00	2.349E-01	1.084E+00
1.037E+04	1.611E-02	3.221E-04	2.014E+00	1.162E-01	1.002E+00

Unknown solution

Solution